

Statistics II 2024/2025 project

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This project aims to develop a model to predict whether a patient has a cardiovascular disease. For developing a model, the dataset “Cardiovascular_Disease_Dataset.csv” was used. The description of the dataset is presented in Table 1. The target variable denotes the presence/absence of the cardiovascular disease in the patient (0 encodes healthy patient, 1 encodes patient with cardiovascular disease).

	Type	Mean	Min	Max	Std	Median	Unique Values	Value Counts	Mode
patientid	Numerical	5.048704e+06	103368.0	9990855.0	2.895905e+06	4952508.5	NaN	NaN	NaN
age	Numerical	4.924200e+01	20.0	80.0	1.786473e+01	49.0	NaN	NaN	NaN
gender	Categorical	NaN	NaN	NaN	NaN	NaN	[1, 0]	{1: 765, 0: 235}	1.0
chestpain	Categorical	NaN	NaN	NaN	NaN	NaN	[2, 0, 1, 3]	{0: 420, 2: 312, 1: 224, 3: 44}	0.0
restingBP	Numerical	1.517470e+02	94.0	200.0	2.996523e+01	147.0	NaN	NaN	NaN
serumcholesterol	Numerical	3.114470e+02	0.0	602.0	1.324438e+02	318.0	NaN	NaN	NaN
fastingbloodsugar	Categorical	NaN	NaN	NaN	NaN	NaN	[0, 1]	{0: 704, 1: 296}	0.0
restingrelectro	Categorical	NaN	NaN	NaN	NaN	NaN	[1, 0, 2]	{0: 454, 1: 344, 2: 202}	0.0
maxheartrate	Numerical	1.454770e+02	71.0	202.0	3.419027e+01	146.0	NaN	NaN	NaN
exerciseangia	Categorical	NaN	NaN	NaN	NaN	NaN	[0, 1]	{0: 502, 1: 498}	0.0
oldpeak	Numerical	2.707700e+00	0.0	6.2	1.720753e+00	2.4	NaN	NaN	NaN
slope	Categorical	NaN	NaN	NaN	NaN	NaN	[3, 1, 2, 0]	{2: 322, 1: 299, 3: 199, 0: 180}	2.0
noofmajorvessels	Categorical	NaN	NaN	NaN	NaN	NaN	[3, 1, 0, 2]	{1: 344, 0: 275, 2: 265, 3: 116}	1.0
target	Categorical	NaN	NaN	NaN	NaN	NaN	[1, 0]	{1: 580, 0: 420}	1.0

Table 1. Predictors and target statistics summary of the dataset “Cardiovascular_Disease_Dataset.csv”.

Categorical variables: gender, chestpain, fastingbloodsugar, restingrelectro, exerciseangia, slope, noofmajorvessels.

Numerical variables: age, restingBP, serumcholesterol, maxheartrate, oldpeak.

Methodology

Principal Component Analysis (PCA)

PCA was used to identify which variables contribute to the most variance in the dataset. The aim was to see whether sick and healthy patients form distinguishable clusters in the reduced-dimensional space. Principal components were also used for modelling.

Factor Analysis (FA)

FA was used to identify health-related factors (e.g., those connected to heart function or blood circulation) that can be used for predictive modelling. It also helped to reduce redundancy in the data and focus on the smaller set of variables.

One-way ANOVA

One-way ANOVA was used to identify whether healthy individuals had mean of principal component 1 significantly different from that of a group of sick individuals.

Logistic regression

Logistic regression was used as a baseline model for classification. It is easy to implement and interpret in binary classification problems. Logistic regression models the relationship between predictors and independent variable using sigmoid function:

$$p = \frac{1}{1 + \exp(-(W^T X + b))}$$

Usually, the threshold for classification is set to 0.5, and if $p > 0.5$, logistic regression predicts class 1, and if $p < 0.5$, logistic regression predicts class 0. The model finds the weights W and b by maximising the likelihood of observing the actual data.

Data analysis

Please refer to the file “Statistics II Data Analysis Dynko.ipynb” for the data analysis part.

Summary

The Receiver Operating Characteristic (ROC) curves were plotted to compare the models. The ROC curves show a trade-off between the true positive rate, TPR (the number of correctly classified sick people out of all sick people) and the false positive rate, FPR (the number of healthy people that were classified as sick). Each ROC's Area Under the Curve (AUC) is calculated to compare models. The dashed line indicates the model with equal TPR and FPR, meaning that the model has no predictive power ($AUC=0.5$). Curves closer to the top-left corner indicate better models with high TPR but low FPR. The summary of model performance is presented in Table. 2, and the ROC curves are shown in Figure 1.

Model	Accuracy	True positive rate	False positive rate	AUC
Logistic regression	0.97	0.97 (113 out of 116)	0.036 (3 out of 84)	0.9969
Logistic regression with factors from FA	0.95	0.96 (112 out of 117)	0.06 (5 out of 83)	0.9904
Logistic regression with first principal component of PCA	0.75	0.76 (89 out of 117)	0.27 (22 out of 83)	0.8424
Logistic regression with first principal component of PCA + four categorical variables	0.96	0.96 (112 out of 117)	0.036 (3 out of 83)	0.9890

Table 2. Comparison of four models created during data analysis based on models' accuracy, true positive rate, false positive rate and AUC.

For creating the models, factor analysis and principal components analysis were applied to identify common factors/components that can help differentiate between healthy and sick patients. The resulting components/factors were then used as data for logistic regression models. Additionally, one logistic regression model was done without factors and principal components but only with variables that were correlated with the target variable (presence of disease) and had high loadings in FA and PCA. These variables were restingBP, serumcholesterol, maxheartrate, oldpeak, slope, chestpain, noofmajorvessels, target, restingelectro.

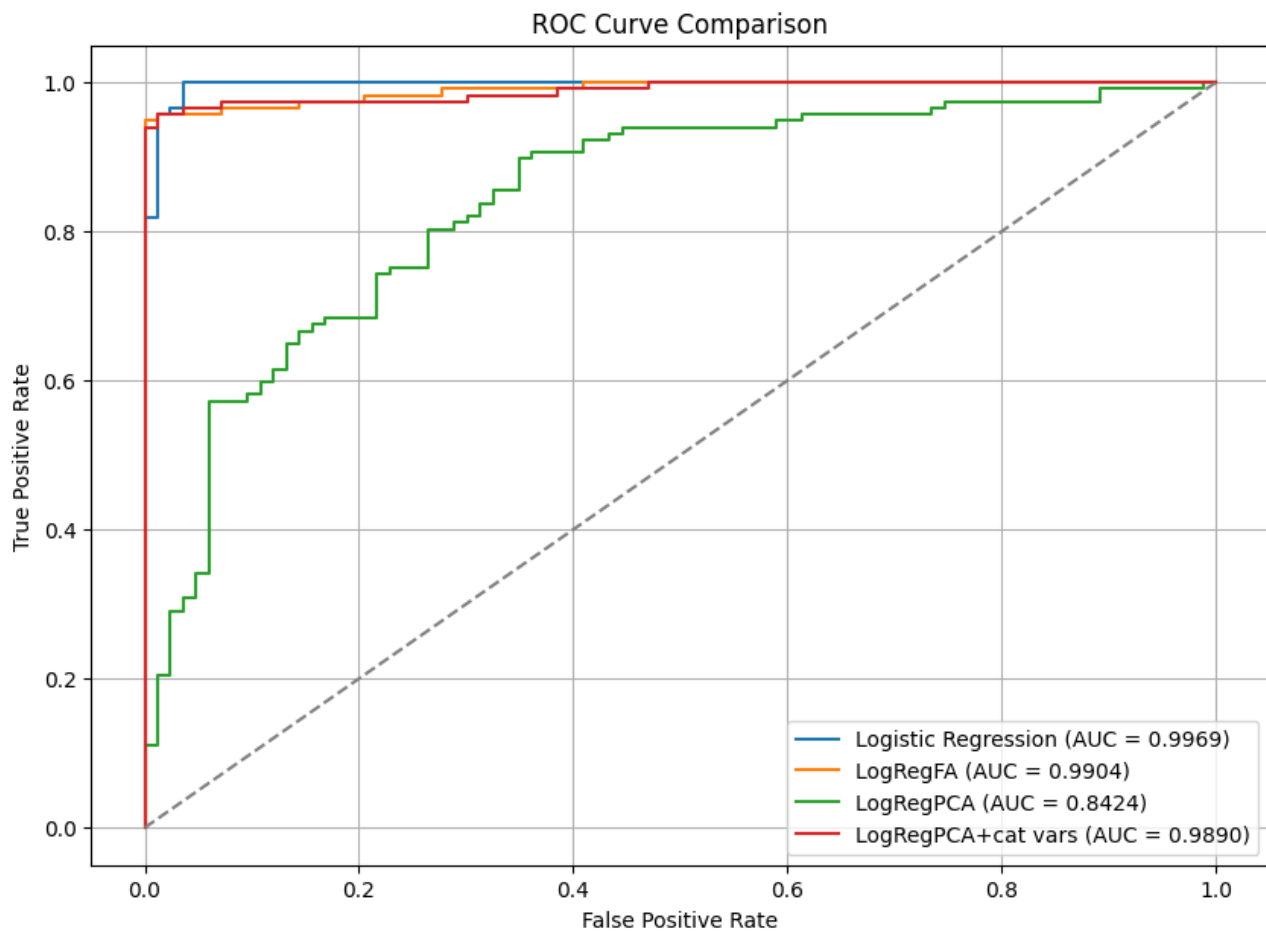


Figure 1. ROC curves for four models that were created during data analysis.

Among the four models, logistic regression using selected variables outperformed all other models with an accuracy of 0.97 and the lowest false negatives (misclassifications of sick people) or false positives (misclassifications of healthy people). It also has the biggest AUC compared to other models (AUC=0.9969). The factor analysis-based model performed better than the PCA-based model. PCA discarded too much variance, resulting in a weaker predictive model. However, when categorical variables were included, the PCA-based model improved significantly, highlighting the importance of retaining categorical features in dimensionality reduction techniques.